

TREMORTRACE · REQUIREMENTS

TremorTrace requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Design for construction (TRT-DDR-003): R6 and R10 figures; budget as a value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Apply TRT-DDR-001 decisions: R5 redefined with a raw-snippet reserve, R11 names the LED and web Bluetooth page, wrist choice; status from TRT-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

TremorTrace requirements

These requirements were checked by calculation in TRT-CAL-001 (TRL 3). Nine are met by calculation or design review and R7 and R8 cannot be verified on paper; none is not met. Changes in v0.3 follow Amish's decisions of 2026-09-25 (TRT-DDR-001). In v0.4, R6 moves from at risk to met because the textile strap recommended at TRL 3 was accepted (TRT-DDR-002), and R4 names the 50 mA charge setting. In v0.5 the mass and cost figures follow the constructable design (TRT-DDR-003) and the budget is treated as a value-engineering target; no status changes.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status (TRT-CAL-001)
R1	Measure tremor frequency across the clinically relevant band	3 to 15 Hz, resolution 0.2 Hz or better	Calculation of sampling and window length; later bench shaker test	Met, if the firmware corrects the sample rate against the crystal
R2	Measure tremor amplitude	Acceleration RMS and angular velocity RMS in the tremor band, reported per window	Calculation; later comparison against a reference sensor	Met
R3	Sample motion fast enough	6-axis IMU at 100 Hz or more	Datasheet and firmware configuration	Met
R4	Run a full waking day and more	3 days or more between charges at 16 h per day of wear; recharge at the 50 mA setting	Power budget calculation	Met, 4.5 to 6.3 days, if the firmware sleeps
R5	Store a week of results on the band	7 days of 10 s window summaries at 16 h of wear per day without a phone, alongside a reserved area for on-demand raw snippets	Storage calculation	Met, 9.7 days
R6	Comfortable for all-day wear	Mass 30 g or less including strap; pod no larger than 45 x 35 x 14 mm	Massing model, then weighing	Met: 26.2 g with textile strap (TRT-CAL-001 v0.3); size met
R7	Skin-safe and wearable	Silicone or textile strap; no exposed electronics; splash resistant (IP54 target)	Design review	Not verifiable at TRL 3
R8	Produce a daily tremor profile	Open notebook turning window summaries into a daily chart of tremor time, frequency and amplitude	Run on sample data	Not verifiable at TRL 3
R9	Protect health data	Data stored locally and exported only by the owner; no cloud dependency	Design review	Met (design review)
R10	Low cost and buildable	Parts cost reported against the USD 150 value-engineering target; no custom PCB required for the first build	Priced BOM	Met: USD 43.69, USD 106.31 under the target
R11	Clear status	Status LED on the band and the web Bluetooth page show battery, recording and sync state	Design review	Met (design review)

■ Assumptions

- Parkinsonian rest tremor is typically about 4 to 6 Hz and essential tremor about 4 to 12 Hz; a 3 to 15 Hz band covers both with margin.
- Wear time of 16 h per day and nightly charging is acceptable, but a multi-day battery is preferred so a missed charge does not lose a day.

- One band is worn on the most affected wrist (decided by Amish, 2026-09-25, TRT-DDR-001); both wrists are later work.
- The mass limit in R6 includes the strap. The strap is a woven textile strap of about 8 g (TRT-DDR-002) and remains the least certain item in the mass estimate.

Safety: TremorTrace is a research and educational prototype, not a medical device. It carries a lithium polymer cell against the skin; the cell must be protected and must never be charged while the band is worn.